Chapter 1. Real Numbers and Limits of Sequences

Field and order axioms, completeness, limits, monotone sequences, limsup and liminf, countability

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Analysis rests on a precise description of the real numbers. We take $\mathbb{R}$ as given by the field axioms, the order axioms and the completeness axiom, and derive everything else from them; Appendix A constructs a field satisfying the axioms. Book numbers are given in brackets.

1.1 Field and order axioms

Definition (Ordered field)

A **field** is a set F with operations $+$ and $\cdot$, both associative and commutative, with $\cdot$ distributive over $+$, identities $0 \neq 1$, additive inverses $-a$, and multiplicative inverses a^{-1} for $a \neq 0$ (axioms (F1)–(F9)). A field is **ordered** if there is a subset P (the positive elements) with (O1) $a, b \in P \Rightarrow a+b \in P$; (O2) $a, b \in P \Rightarrow ab \in P$; (O3) for each a exactly one of $a \in P$, $a = 0$, $-a \in P$ holds. We write $a < b$ (or $b > a$) if $b - a \in P$, and $a \leq b$ if $a < b$ or $a = b$.

Proposition 1.1 (1.1.1)

In a field:
(a) $-(-a) = a$, and $(a^{-1})^{-1} = a$ for $a \neq 0$;
(b) $a + b = a + c \Rightarrow b = c$, and $ab = ac$, $a \neq 0 \Rightarrow b = c$;
(c) $ab = 0 \iff a = 0$ or $b = 0$;
(d) $a \cdot 0 = 0$, $(-a)b = -(ab)$, $(-a)(-b) = ab$;
(e) the identities and inverses are unique.

Proof. (e) If $0, 0'$ are additive identities, $0 = 0 + 0' = 0'$; if b, b' are inverses of a, $b = b + (a + b') = (b + a) + b' = b'$; likewise for $\cdot$.

(b) Add $-a$ to both sides (multiply by a^{-1}).

(a) a and $-(-a)$ are both additive inverses of $-a$; use (e).

(d) $a \cdot 0 + a \cdot 0 = a(0 + 0) = a \cdot 0 = a \cdot 0 + 0$, so $a \cdot 0 = 0$ by (b); $ab + (-a)b = (a - a)b = 0$ gives $(-a)b = -(ab)$, and then $(-a)(-b) = -(a(-b)) = -(-(ab)) = ab$.

(c) If $ab = 0$ and $a \neq 0$, then $b = a^{-1}ab = 0$. ∎

Proposition 1.2 (1.1.2)

In an ordered field:
(a) $a \leq b$ and $b \leq a$ imply $a = b$;
(b) $a < b$, $b < c$ imply $a < c$;
(c) $a + b < a + c \iff b < c$;
(d) if $a > 0$ then $b < c \iff ab < ac$, and if $a < 0$ then $b < c \iff ab > ac$;
(e) $a^2 > 0$ for $a \neq 0$, in particular $1 > 0$;
(f) $a > 0 \Rightarrow a^{-1} > 0$, and $0 < a < b \Rightarrow 0 < b^{-1} < a^{-1}$;
(g) $a < b$, $c < d \Rightarrow a + c < b + d$;
(h) for $a, b \geq 0$: $a < b \iff a^2 < b^2$, and $a^2 = b^2 \Rightarrow a = b$.

Proof. (a) If $a \neq b$, both $b - a$ and $a - b = -(b - a)$ would be in P, against (O3).

(b) $c - a = (c - b) + (b - a) \in P$.

(c) $(a + c) - (a + b) = c - b$.

(d) $ac - ab = a(c - b)$; for $a > 0$ use (O2) and trichotomy, for $a < 0$ note $-a > 0$.

(e) If $a > 0$, $a^2 \in P$; if $a < 0$, $a^2 = (-a)^2 \in P$; and $1 = 1^2$.

(f) If $a^{-1} \le 0$ then $1 = aa^{-1} \le 0$; multiply $a < b$ by $a^{-1}b^{-1} > 0$.

(g) $(b+d)-(a+c) = (b-a)+(d-c)$.

(h) $b^2 - a^2 = (b-a)(b+a)$ and $b+a > 0$ when $a, b \ge 0$ are not both 0. ■

$\mathbb{Q}$ and $\mathbb{R}$ are ordered fields. $\mathbb{C}$ is not: in an ordered field $-1 < 0 < 1 = 1^2$, but $i^2 = -1$ would have to be positive. No finite field is ordered, since $1, 1+1, 1+1+1, \ldots$ are strictly increasing.

The **absolute value** is $|a| = a$ for $a \ge 0$ and $|a| = -a$ for $a < 0$.

Proposition 1.3 (1.1.3)

(a) $|a| \ge 0$, and $|a| = 0 \iff a = 0$;
(b) $|ab| = |a||b|$;
(c) for $b \ge 0$, $|a| \le b \iff -b \le a \le b$;
(d) $|a+b| \le |a| + |b|$ (triangle inequality);
(e) $||a| - |b|| \le |a-b|$.

Proof. (a), (b) are case checks.

(c) If $a \ge 0$, $|a| \le b \iff a \le b$, and $-b \le 0 \le a$ automatically; similarly for $a < 0$.

(d) From $-|a| \le a \le |a|$ and the same for b, $-(|a|+|b|) \le a+b \le |a|+|b|$; apply (c).

(e) $|a| = |(a-b)+b| \le |a-b| + |b|$, and symmetrically. ■

1.2 The completeness axiom

$S \subset \mathbb{R}$ is **bounded above** if it has an **upper bound** u ($s \le u$ for all $s \in S$). A **least upper bound** (**supremum**) $\sup S$ is an upper bound smaller than every other upper bound; it is unique if it exists. Lower bounds and $\inf S$ are defined dually.

Definition (Completeness axiom)

(C) Every nonempty subset of $\mathbb{R}$ that is bounded above has a least upper bound.

Proposition 1.4 (1.2.1)

In an ordered field the following are equivalent:
(a) every nonempty set bounded above has a supremum;
(b) every nonempty set bounded below has an infimum.

Proof. (a)⇒(b): let $S \ne \emptyset$ be bounded below and L the set of lower bounds of S; $L \ne \emptyset$, and every $s \in S$ is an upper bound of L. Let $a = \sup L$. Each $s \in S$ is an upper bound of L, so $a \le s$: $a \in L$. If $b \in L$ then $b \le a$. So $a = \inf S$.

(b)⇒(a) is symmetric, or apply (b) to $-S = \{-s\}$. ■

A useful reformulation: $u = \sup S$ iff u is an upper bound and for every $\epsilon > 0$ there is $s \in S$ with $s > u - \epsilon$.

Proposition 1.5 (1.2.2)

$\mathbb{N}$ is not bounded above.

Proof. Otherwise $a = \sup \mathbb{N}$ exists; $a - 1$ is not an upper bound, so $a - 1 < n$ for some $n \in \mathbb{N}$, and then $n + 1 > a$ with $n + 1 \in \mathbb{N}$, a contradiction. ■

Corollary 1.6 (1.2.3)

$\inf\{1/n : n \in \mathbb{N}\} = 0$; that is, for every $\epsilon > 0$ there is $n \in \mathbb{N}$ with $1/n < \epsilon$.

Proof. 0 is a lower bound. If $\epsilon > 0$ were a lower bound, then $n \leq 1/\epsilon$ for all n, contradicting Proposition 1.5. ■

Corollary 1.7 (1.2.4, Archimedean property)

For $a > 0$ and $b \in \mathbb{R}$ there is $n \in \mathbb{N}$ with $na > b$.

Proof. By Proposition 1.5 choose $n > b/a$. ■

Theorem 1.8 (1.2.5, density of the rationals)

If $a < b$ there is $r \in \mathbb{Q}$ with $a < r < b$.

Proof. First let $0 \leq a < b$. Choose n with $1/n < b - a$ (Corollary 1.6). The set $\{m \in \mathbb{N} : m/n > a\}$ is nonempty (Corollary 1.7); let m be its least element. Then $(m-1)/n \leq a$ (if $m = 1$ this holds as $a \geq 0$), so $a < m/n \leq a + 1/n < b$. If $a < 0$, choose $k \in \mathbb{N}$ with $k > -a$, find r between $a + k \geq 0$ and $b + k$, and take $r - k$. ■

The irrationals are also dense: apply the theorem to $a/\sqrt{2} < b/\sqrt{2}$ (using Proposition 1.9).

Proposition 1.9 (1.2.6)

For each $a > 0$ there is a unique $b > 0$ with $b^2 = a$.

Proof. Uniqueness: Proposition 1.2(h).

Existence: $S = \{x > 0 : x^2 < a\}$ is nonempty (it contains $t = a/(1 + a)$, since $t < 1$ and $t < a$ give $t^2 < t < a$) and bounded above by $1 + a$ (if $x > 1 + a$ then $x^2 > x > a$). Let $b = \sup S > 0$. If $b^2 < a$, for $0 < h < 1$ we have $(b+h)^2 \leq b^2 + h(2b+1) < a$ once $h < (a - b^2)/(2b+1)$, so $b + h \in S$, a contradiction. If $b^2 > a$, for $0 < h < b$, $(b - h)^2 > b^2 - 2bh > a$ once $h < (b^2 - a)/2b$; then $b - h$ is an upper bound of S (if $x \geq b - h$ then $x^2 \geq (b - h)^2 > a$), a contradiction. So $b^2 = a$. ■

In particular $\mathbb{Q}$ does not satisfy (C): $\{x \in \mathbb{Q} : x > 0, x^2 < 2\}$ has no rational supremum, as $\sqrt{2}$ is irrational.

Theorem 1.10 (1.2.7, uniqueness of the real field)

Let F and G be ordered fields satisfying the completeness axiom. There is a bijection $f : F \to G$ with $f(a + b) = f(a) + f(b)$, $f(ab) = f(a)f(b)$, and $a < b \iff f(a) < f(b)$.

Proof. Each ordered field contains a copy of $\mathbb{Q}$: $n \mapsto 1 + \cdots + 1$, extended to $\mathbb{Z}$ and to quotients; this map $\mathbb{Q} \to F$ preserves operations and order (Proposition 1.2). Let $\phi : \mathbb{Q}_F \to \mathbb{Q}_G$ be the resulting isomorphism of the rational subfields. Both F and G are Archimedean (Corollary 1.7), so the rationals are dense in each (Theorem 1.8). Define

$$f(x) = \sup\{\phi(r) : r \in \mathbb{Q}_F,\ r < x\}.$$

The set is nonempty and bounded above (by $\phi(s)$ for any rational $s > x$), so $f(x)$ exists. Order: if $x < y$, pick rationals $x < r < s < y$; then $f(x) \leq \phi(r) < \phi(s) \leq f(y)$. Also $f(r) = \phi(r)$

for rational r (density). So f is strictly increasing, hence injective. Surjective: for $z \in G$ let $x = \sup\{r \in \mathbb{Q}_F : \phi(r) < z\}$; density in G and strict monotonicity show $f(x) = z$. Additivity: for rationals $r < x$, $s < y$ we have $r + s < x + y$, so $\phi(r) + \phi(s) \le f(x + y)$, and taking suprema $f(x) + f(y) \le f(x + y)$; applying this to $-x, -y$ (note $f(-x) = -f(x)$, from the definition and density) gives equality. Multiplicativity for $x, y > 0$: products rs of positive rationals $r < x$, $s < y$ are exactly cofinal among rationals below xy, giving $f(xy) = f(x)f(y)$; extend to all signs using $f(-x) = -f(x)$. ∎

So the axioms determine $\mathbb{R}$ up to isomorphism.

Proposition 1.11 (1.2.8)

For nonempty bounded $S, T \subset \mathbb{R}$, $\sup(S+T) = \sup S + \sup T$ and $\inf(S+T) = \inf S + \inf T$, where $S + T = \{s + t\}$. If S, T consist of positive numbers, $\sup(ST) = \sup S \cdot \sup T$ and $\inf(ST) = \inf S \cdot \inf T$.

Proof. $\sup S + \sup T$ is an upper bound of $S+T$. Given $\epsilon > 0$ choose $s > \sup S - \epsilon/2$, $t > \sup T - \epsilon/2$; then $s + t > \sup S + \sup T - \epsilon$. For products, $\sup S \sup T$ is an upper bound of ST; with $a = \sup S$, $b = \sup T$ and $0 < \delta < \min(a, b)$, choose $s > a - \delta$, $t > b - \delta$, so $st > (a-\delta)(b-\delta) = ab - \delta(a+b-\delta)$, which is arbitrarily close to ab. For infima, $\inf S \inf T$ is a lower bound of ST, and with $s < \inf S + \delta$, $t < \inf T + \delta$ we get $st < \inf S \inf T + \delta(\inf S + \inf T + \delta)$. ∎

1.3 Limits of sequences

(x_n) **converges** to x, $\lim x_n = x$, if for every $\epsilon > 0$ there is N with $|x_n - x| < \epsilon$ for all $n \ge N$.

Proposition 1.12 (1.3.1)

Limits are unique.

Proof. If $x_n \to x$, $x_n \to y$ and $x \ne y$, take $\epsilon = |x - y|/2$ and $n \ge \max(N_1, N_2)$: $|x - y| \le |x - x_n| + |x_n - y| < 2\epsilon = |x - y|$. ∎

Proposition 1.13 (1.3.2)

Every convergent sequence is bounded.

Proof. If $x_n \to x$, choose N with $|x_n - x| < 1$ for $n \ge N$; then $|x_n| \le \max\{|x_1|, \dots, |x_{N-1}|, |x| + 1\}$. ∎

Theorem 1.14 (1.3.3, limit laws)

If $x_n \to x$ and $y_n \to y$, then (a) $x_n + y_n \to x + y$;
(b) $ax_n \to ax$;
(c) $x_n y_n \to xy$;
(d) if $y \ne 0$, then $y_n \ne 0$ for large n and $x_n/y_n \to x/y$;
(e) if $x_n \le y_n$ for all large n, then $x \le y$;
(f) (squeeze) if $x_n \le z_n \le y_n$ and $x = y$, then $z_n \to x$.

Proof. (a) $|(x_n + y_n) - (x + y)| \le |x_n - x| + |y_n - y|$; use $\epsilon/2$ for each.

(b) is a special case of (c).

(c) With M bounding $|x_n|$ (Proposition 1.13), $|x_n y_n - xy| \le |x_n||y_n - y| + |y||x_n - x| \le M|y_n - y| + |y||x_n - x|$.

(d) For large n, $|y_n| > |y|/2$, and $|1/y_n - 1/y| = |y - y_n|/(|y_n||y|) \le 2|y_n - y|/|y|^2$; combine with (c).

(e) If $x > y$, take $\epsilon = (x - y)/2$; for large n, $x_n > x - \epsilon = y + \epsilon > y_n$.

(f) For large n, $x - \epsilon < x_n \le z_n \le y_n < x + \epsilon$. ■

Example 1.1

(a) $1/n \to 0$ by Corollary 1.6.
(b) For $|r| < 1$, $r^n \to 0$: write $1/|r| = 1 + h$, $h > 0$; then $(1 + h)^n \ge 1 + nh$ (Bernoulli), so $|r|^n \le 1/(1 + nh)$.
(c) $n^{1/n} \to 1$: with $n^{1/n} = 1 + h_n$, $n \ge \binom{n}{2} h_n^2$, so $0 \le h_n \le \sqrt{2/(n-1)}$.

1.4 Monotone sequences, limsup and liminf

Theorem 1.15 (1.4.1, monotone convergence)

A bounded monotone sequence of real numbers converges.

Proof. Let (x_n) be increasing and bounded, and $x = \sup\{x_n\}$ (axiom (C)). Given ϵ, some $x_N > x - \epsilon$, and then $x - \epsilon < x_N \le x_n \le x$ for $n \ge N$. For decreasing sequences use inf. ■

Corollary 1.16 (1.4.2)

A series of nonnegative terms whose partial sums are bounded converges.

Proof. The partial sums form a bounded increasing sequence. ■

Example 1.2

$(1 + 1/n)^n$ is increasing and bounded by 3 (binomial expansion), so it converges to a number e. $\sum 1/n^p$ converges for $p > 1$: grouping terms in blocks $2^k \le n < 2^{k+1}$, the partial sums are at most $\sum_k 2^{k(1-p)} < \infty$. $\sum 1/n$ diverges: the block sums are at least $\frac{1}{2}$.

For a bounded sequence (x_n), the sequences $\sup_{k \ge n} x_k$ and $\inf_{k \ge n} x_k$ are decreasing and increasing, so they converge (Theorem 1.15); define

$$\limsup_n x_n = \lim_n \sup_{k \ge n} x_k = \inf_n \sup_{k \ge n} x_k, \qquad \liminf_n x_n = \lim_n \inf_{k \ge n} x_k = \sup_n \inf_{k \ge n} x_k.$$

Clearly $\liminf x_n \le \limsup x_n$, and $\liminf x_n = -\limsup(-x_n)$.

Proposition 1.17 (1.4.3)

For bounded (x_n), $a = \limsup x_n$ iff (a) for every $\epsilon > 0$ there is N with $x_n < a + \epsilon$ for all $n \ge N$, and (b) for every $\epsilon > 0$, $x_n > a - \epsilon$ for infinitely many n.

Proof. Let $s_n = \sup_{k \ge n} x_k \downarrow a$.

(a) Choose N with $s_N < a + \epsilon$; then $x_n \le s_N$ for $n \ge N$.

(b) For each N, $s_N \geq a > a-\epsilon$, so some $k \geq N$ has $x_k > a-\epsilon$. Conversely, if a satisfies (a) and (b), then by (a) $s_n \leq a+\epsilon$ for large n, so $\limsup \leq a+\epsilon$; by (b) $s_n > a-\epsilon$ for all n, so $\limsup \geq a-\epsilon$. As ϵ is arbitrary, $\limsup x_n = a$. ■

Proposition 1.18 (1.4.4)

For bounded (x_n), $a = \liminf x_n$ iff (a) for every $\epsilon > 0$ there is N with $x_n > a-\epsilon$ for all $n \geq N$, and (b) for every $\epsilon > 0$, $x_n < a+\epsilon$ for infinitely many n.

Proof. Apply Proposition 1.17 to $(-x_n)$. ■

Theorem 1.19 (1.4.5)

If $x_n \to a$, then $\limsup x_n = \liminf x_n = a$. Conversely, if (x_n) is bounded and $\limsup x_n = \liminf x_n = a$, then $x_n \to a$.

Proof. If $x_n \to a$, then a satisfies conditions (a) and (b) of both propositions. Conversely, part (a) of Propositions 1.17 and 1.18 give $a-\epsilon < x_n < a+\epsilon$ for all large n. ■

Theorem 1.20 (1.4.6, nested interval theorem)

If $I_n = [a_n, b_n]$ are bounded closed intervals with $I_{n+1} \subset I_n$ for all n, then $\bigcap_n I_n \neq \emptyset$. If moreover $b_n - a_n \to 0$, the intersection is a single point.

Proof. (a_n) is increasing and bounded by b_1, so $a_n \to a = \sup a_n$; for all m, n, $a_n \leq a_{\max(m,n)} \leq b_{\max(m,n)} \leq b_m$, so $a \leq b_m$. Thus $a_m \leq a \leq b_m$ for all m. If x, y both lie in every I_n, $|x-y| \leq b_n - a_n \to 0$. ■

Conversely, the nested interval theorem together with the Archimedean property implies the completeness axiom: for S bounded above, bisect an interval $[s_0, u_0]$ ($s_0 \in S$, u_0 an upper bound) repeatedly, keeping the half whose right end point is an upper bound and whose interval meets S; the common point of the intervals is $\sup S$. Similarly, Theorem 1.15 is equivalent to (C).

1.5 Countable and uncountable sets

A set is **countable** if it is finite or in bijection with $\mathbb{N}$.

Proposition 1.21 (countable unions)

Subsets of countable sets are countable; $\mathbb{N} \times \mathbb{N}$ is countable; a countable union of countable sets is countable. Hence $\mathbb{Z}$ and $\mathbb{Q}$ are countable.

Proof. An infinite $A \subset \mathbb{N}$ is enumerated increasingly by recursion ($a_1 = \min A$, $a_{k+1} = \min(A - \{a_1, \ldots, a_k\})$). $(m, n) \mapsto 2^m 3^n$ injects $\mathbb{N}\times\mathbb{N}$ into $\mathbb{N}$. If A_k are countable with surjections $f_k : \mathbb{N} \to A_k$ (for nonempty A_k), then $(k, n) \mapsto f_k(n)$ maps $\mathbb{N} \times \mathbb{N}$ onto $\bigcup A_k$, and a set that is the image of a countable set is countable (choose least preimages). $\mathbb{Q}$ is the image of $\mathbb{Z} \times \mathbb{N}$ under $(p, q) \mapsto p/q$. ■

Theorem 1.22 (Cantor)

$\mathbb{R}$ is uncountable.

Proof. Let $x_1, x_2, \ldots$ be any sequence of reals. Choose a closed interval I_1 of positive length not containing x_1, then $I_2 \subset I_1$ not containing x_2, and so on. By Theorem 1.20 some $x \in \bigcap I_n$, and $x \neq x_n$ for every n. ■

Consequently the irrational numbers are uncountable, and every interval of positive length is uncountable.